

Mukund Mahakalkar

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SUMMARY

Computer Science undergraduate at IIT Bhopal with hands-on experience developing modules for a healthcare **Revenue Cycle Management** platform using React.js, Node.js, TypeScript, and AWS during a 4-month internship. Independently built and deployed 3 full-stack applications with Next.js, Docker, PostgreSQL, MongoDB, and Redis, integrating OpenAI and Gemini APIs. **700+ LeetCode** problems solved (Knight, rating **1862**).

EDUCATION

Indian Institute of Information Technology (IIIT)

Bachelor of Technology in Computer Science and Engineering; GPA: 8.62/10

Bhopal, India

June 2023 – June 2027

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems

EXPERIENCE

QuickIntell

Remote

Full Stack Web Developer Intern

December 2025 – March 2026

- Built the Scribe and Medical Coding modules for a US healthcare Revenue Cycle Management (RCM) platform using React.js, TypeScript, Node.js, and Prisma ORM; automated audio transcription into structured SOAP notes with ICD-10/CPT code generation via the OpenAI API, cutting manual coding time by **35%**.
- Designed an AI inference pipeline that routes clinical PDFs through Claude Haiku and escalates low-confidence cases to Claude Opus with a safety-validation step, built on asynchronous Amazon SQS/S3 processing to handle multiple billing-code categories.
- Built a shared component library with Tailwind CSS and ShadCN UI that cut frontend development time by **40%**; wrote unit tests with Jest and connected Prisma ORM queries to paginated RESTful API endpoints.

PROJECTS

ThreadStream | Next.js, TypeScript, Socket.IO, PostgreSQL, Clerk Auth, Cloudinary, Zod

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- Built a **real-time platform** combining a community forum (threads, replies, likes), direct messaging, and group chat rooms; added Cloudinary image sharing and secured all REST endpoints with Clerk authentication.
- Implemented **Socket.IO** events for direct/group messaging, typing indicators, message edit and delete, read receipts, and online presence, using user- and room-scoped WebSocket channels for event routing.
- Designed a normalized PostgreSQL schema with composite indexes and versioned migrations; added rate limiting, Helmet security headers, Zod validation, and a **GitHub Actions CI/CD** pipeline for automated TypeScript checks.

ShopEasy | MongoDB, Express.js, React.js, Node.js, Redux Toolkit, Gemini API, Clerk Auth, PayPal SDK

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- Built an e-commerce application with a Redux Toolkit shopping cart, MongoDB-backed stock validation, and product filtering/search; deployed on Vercel and Render.
- Implemented an Admin Dashboard with **Role-Based Access Control (RBAC)** for order management and product CRUD, using Multer and Cloudinary for image handling.
- Secured transactions with the **PayPal SDK**, Clerk OAuth, JWT, and bcrypt hashing.
- Integrated the **Gemini API** for an AI shopping assistant, visual product search, and review sentiment analysis, plus AI-generated sales insights on the admin dashboard.

MicroMedia | Node.js, Express.js, RabbitMQ, Redis, MongoDB, Cloudinary, Docker

[GitHub](#)

- Decomposed a monolith into **5 microservices** (Identity, Post, Media, Search, API Gateway); routed requests through an Express HTTP Proxy gateway, validated JWTs, and enforced Redis rate limiting across all REST endpoints.
- Built an **event-driven architecture** on a RabbitMQ topic exchange with message acknowledgement; creation/deletion events are consumed by the Media Service (cleans up unused Cloudinary assets) and the Search Service (MongoDB full-text indexing).
- Applied Redis caching with time-based invalidation, Argon2 password hashing, Joi schema validation, and DDoS rate limiting; containerized all **7 components** (5 services plus RabbitMQ and Redis) with Docker Compose and health checks.

TECHNICAL SKILLS

Languages:	C++, JavaScript (ES6+), TypeScript, HTML, CSS
Frontend:	React.js, Next.js, Redux Toolkit, Tailwind CSS, ShadCN UI, Jest
Backend:	Node.js, Express.js, Microservices Architecture, RabbitMQ, Redis, REST APIs, WebSockets
Cloud and DevOps:	Amazon Web Services (S3, SQS), Docker, Vercel, Render, CI/CD Pipelines
Databases:	MongoDB, MySQL, PostgreSQL, Prisma ORM
Developer Tools:	Git, GitHub, VS Code, Postman, Cloudinary, Linux
Core Concepts:	DSA, OOP, Database Management Systems, Operating Systems, System Design

ACHIEVEMENTS

- Competitive Programming:** LeetCode Knight (rating 1862, top 5%, **700+** solved); CodeChef 3-Star (rating 1633), best global rank **232** (CodeChef Starters 249); Codeforces Pupil (rating 1201).
- Adobe GenSolve Hackathon:** Ranked in the top 10% nationally, advancing to Round 2 from over **250,000** participants.
- NTSE Scholar 2021:** Recognized as top 0.2% nationally by NCERT for academic excellence.